Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

When ethanol and acetic acid were mixed together in equilimolecular proportion 66.6% are converted into ethyl acetate. Calculate $\mathrm{K_C}$. Also calculate quantity of ester produced if one mole of acetic acid is treated with 0.5 mole and 4 mole of alcohol respectively.

(A) $4, 0.93, 0.43$

(B) $0.93, 4, 0.43$

(C) $0.43, 0.93, 4$

(D) $4, 0.43, 0.93$

Q.2

Two moles of ammonia was introduced in an evacuated vessel of 1 litre capacity. At high temperature the gas undergoes partial dissociation according to the equation :

$$\mathrm{NH_3(\ g) \rightleftharpoons N_2(\ g) + 3H_2(\ g)}$$

At equilibrium the concentration of ammonia was found to be 1 mole. What is the value of ' K ' ?

(A) $3/4 = 0.75\ \mathrm{mol^2 \ell^{-2}}$

(B) $3/2 = 1.5\ \mathrm{mol^2 \ell^{-2}}$

(C) $27/16 = 1.7\ \mathrm{mol^2 \ell^{-2}}$

(D) $27/64 = 0.42\ \mathrm{mol^2 \ell^{-2}}$

Q.3

1 mole $\mathrm{N_2}$ and $3\ \mathrm{mol H_2}$ are placed in a closed container at a pressure of $4\ \mathrm{atm}$. The pressure falls to 3 atm at the same temperature when the following equilibrim is attained.

$\mathrm{N_2(g) + 3H_2(\ g) \rightleftharpoons 2NH_3(g)}$. The equilibrium constant $\mathrm{K_p}$ for dissociation of $\mathrm{NH_3}$ is :

(A) $\frac{1}{0.5} \times (1.5)^3\ \mathrm{atm^{-2}}$

(B) $0.5 \times (1.5)^3\ \mathrm{atm^2}$

(C) $\frac{0.5 \times (1.5)^3}{3 \times 3}\ \mathrm{atm^2}$

(D) $\frac{3 \times 3}{0.5 \times (1.5)^3}\ \mathrm{atm^{-2}}$

Q.4

@IITJEE_Advanced

$4.0 \mathrm{gms}$ of hydrogen react with 9.023×10^{23} molecules of chlorine to form HCl gas. The total pressure after the reaction was found to be $700 \mathrm{~mm}$. The partial pressure of HCl will be -

(A) $3900 \mathrm{~mm}$

(B) $600 \mathrm{~mm}$

(C) $700 \mathrm{~mm}$

(D) $350 \mathrm{~mm}$

Q.5

Equilibrium constants are given (in atm) for the following reactions $0^{\circ} \mathrm{C}$
$\mathrm{SrCl_2.6H_2O(s)} \rightleftharpoons \mathrm{SrCl_2 \cdot 2H_2O}(s)+4 \mathrm{H_2O(g)} \quad K_p=5 \times 10^{12}$
$\mathrm{Na_2HPO_4 \cdot 12H_2O(s)} \rightleftharpoons \mathrm{Na_2HPO_4 7H_2O}(s)+5 \mathrm{H_2O(g)}$
$K_p=2.43 \times 10^{13}$
$\mathrm{Na_2SO_4 10H_2O}(s) \rightleftharpoons \mathrm{Na_2SO_4}(s)+10 \mathrm{H_2O(g)} \quad \mathrm{K}_p=1.024 \times 10^{27}$
The vapour pressure of water at $0^{\circ} \mathrm{C}$ is 4.56 torr. Which is the most effective drying agent at $0^{\circ} \mathrm{C}$

(A) $\mathrm{SrCl_2 \cdot 2H_2O}$

(B) $\mathrm{Na_2SO_4}$

(C) $\mathrm{Na_2HPO_4 7H_2O}$

(D) all equally

Q.6

Consider the following equilibrium at $25^{\circ} \mathrm{C}, 2 \mathrm{NO(g)} \rightleftharpoons \mathrm{N_2(~g)+O_2(~g) ; K_1=4 \times 10^{30}}$ and $\mathrm{NO(g)}+\frac{1}{2} \mathrm{Br_2(~g)} \rightleftharpoons \mathrm{NOBr(g) ; K_2=1.4~mol^{-1/2}}$ litre $^{1/2}$
The value of $\mathrm{K_c}$ for the reaction (at the same temperature) $\frac{1}{2} \mathrm{~N_2(~g)}+\frac{1}{2} \mathrm{O_2(~g)}+\frac{1}{2} \mathrm{Br_2(~g)} \rightleftharpoons \mathrm{NOBr(g)}$

(A) 3.5×10^{-31}

(B) 2.8×10^{15}

(C) 7.0×10^{-16}

(D) 5.6×10^{30}

Q.7

A mixture of $\mathrm{NH_3(~g)}$ and $\mathrm{N_2H_4(~g)}$ is placed in a sealed container at $300 \mathrm{~K}$ .The total pressure of the gaseous mixture is $0.5 \mathrm{~atm}$. The container is heated to $1200 \mathrm{~K}$ at which both $\mathrm{NH_3}$ and $\mathrm{N_2H_4}$ decomposed completely according to the following equation .
$\mathrm{2NH_3(~g) \longrightarrow N_2(~g)+3H_2(~g)}$
$\mathrm{N_2H_4(~g) \longrightarrow N_2(~g)+2H_2(~g)}$
After decomposition is complete, the total pressure at $1200 \mathrm{~K}$ is found to be 4.5 atm. The mole $\%$ of $\mathrm{N_2H_4}$ in the initial mixture at $300 \mathrm{~K}$ was

(A) 20%

(B) 25%

@IITJEE_Advanced

(C) 50%

(D) 75%

Q.8

For the reaction, $\mathrm{A}+\mathrm{B} \rightleftharpoons 3\mathrm{C}$ at $25^{\circ}\mathrm{C}$, a $3\ \mathrm{L}$ vessel contains $1, 2, 4$ mole of A, B and C respectively. Predict the direction of reaction -

(A) Forward if $\mathrm{K_C}$ for the reaction is 14

(B) Backward if $\mathrm{K_C}$ for the reaction is 14

(C) Forward if $\mathrm{K_C}$ for the reaction is 9

(D) Backward if $\mathrm{K_C}$ for the reaction is 9

Q.9

When $\mathrm{NaNO_3(\ s)}$ is heated in a closed vessel, oxygen is liberated and $\mathrm{NaNO_2(\ s)}$ is left behind. At equilibrium.

(A) addition of $\mathrm{NaNO_2}$ favours reverse reaction

(B) addition of $\mathrm{NaNO_3}$ favours forward reaction

(C) increasing temperature favours forward reaction

(D) increasing pressure favours reverse reaction

Q.10

If two gases $\mathrm{AB_2}$ and $\mathrm{B_2C}$ are mixed the following equilibria are readily established

$$\mathrm{AB_2(\ g) + B_2C(g) \rightarrow AB_3(\ g) + BC(g)}$$
$$\mathrm{BC(g) + B_2C(g) \rightarrow B_3C_2(\ g)}$$

If the reaction is started only with $\mathrm{AB_2}$ with $\mathrm{B_2C}$, then which of the following is necessarily true at equilibrium

(A) $[\mathrm{AB_3}]_{\mathrm{eq}} = [\mathrm{BC}]_{\mathrm{eq}}$

(B) $[\mathrm{AB_2}]_{\mathrm{eq}} = [\mathrm{B_2C}]_{\mathrm{eq}}$

(C) $[\mathrm{AB_3}]_{\mathrm{eq}} > [\mathrm{B_3C_2}]_{\mathrm{eq}}$

(D) $[\mathrm{AB_3}]_{\mathrm{eq}} > [\mathrm{BC}]_{\mathrm{eq}}$

Q.11

$\mathrm{NH_2COONH_4} \rightleftharpoons \mathrm{NH_3} + \mathrm{CO_2}$

At equilibrium state, which of the following features are correct?

(A) Total pressure is 1.5 times that of partial pressure of $\mathrm{NH_3}$.

(B) K_P value is 4 times that of $\left(P_{\mathrm{CO_2}}\right)^3$.

(C) $P_{\mathrm{NH_3}} = \frac{1}{2} P_{\mathrm{CO_2}}$

@IITJEE_Advanced

(D) CO_2 contributes 33.3% of equilibrium pressure

Q.12

Paragraph:
$2\ A_2 \rightleftharpoons A_4$...(i)
$K_{p_1} = \frac{2}{81}\ atm^{-1}$
$A_2 + 2C \rightleftharpoons A_2C_2$...(ii)

$$A_2C_2 \rightleftharpoons 2AC...(iii)$$

A_2 and C are taken in $3:1$ mole ratio in a closed container of a certain volume at a fixed temperature and above three equilibriums are established simultaneously. for the first reaction is $2/81\ atm^{-1}$. At equilibrium partial pressure of $A_4(g)$ and $AC(g)$ are found to be $1/2\ atm$ each and the total pressure at equilibrium is found to be $27/4\ atm$. Then-

Question:
The partial pressure of A_2C_2 at equilibrium is -

(A) $1/2$

(B) $1/4$

(C) $3/4$

(D) 1

Q.13

Paragraph:
$2\ A_2 \rightleftharpoons A_4$...(i)
$K_{p_1} = \frac{2}{81}\ atm^{-1}$
$A_2 + 2C \rightleftharpoons A_2C_2$...(ii)

$$A_2C_2 \rightleftharpoons 2AC...(iii)$$

A_2 and C are taken in $3:1$ mole ratio in a closed container of a certain volume at a fixed temperature and above three equilibriums are established simultaneously. for the first reaction is $2/81\ atm^{-1}$. At equilibrium partial pressure of $A_4(g)$ and $AC(g)$ are found to be $1/2\ atm$ each and the total pressure at equilibrium is found to be $27/4\ atm$. Then-

Question:
The mole ratio of gases A_2 and AC at equilibrium is -

(A) $9/2$

(B) $7/2$

(C) 8

(D) 9

@IITJEE_Advanced

Q.14

Paragraph:

$2\ A_2 \rightleftharpoons A_4$...(i)

$K_{p_1} = \frac{2}{81}\ atm^{-1}$

$A_2 + 2C \rightleftharpoons A_2C_2$...(ii)

$$A_2C_2 \rightleftharpoons 2AC...(iii)$$

A_2 and C are taken in $3:1$ mole ratio in a closed container of a certain volume at a fixed temperature and above three equilibriums are established simultaneously. for the first reaction is $2/81\ atm^{-1}$. At equilibrium partial pressure of $A_4(g)$ and $AC(g)$ are found to be $1/2\ atm$ each and the total pressure at equilibrium is found to be $27/4\ atm$. Then-

Question:

Equilibrium constant K_p for the reaction : $2AC \rightleftharpoons A_2C_2$ is -

(A) $3/4$

(B) 3

(C) 4

(D) $1/3$

Q.15

Paragraph:

Following equilibria are established on mixing two gases A_2 and C.

(i) $3\ A_2(g) \rightleftharpoons A_6(g); K_p = 1.6\ atm^{-2}$

(ii) $A_2(g) + C(g) \rightleftharpoons A_2C(g)$

A_2 and C are mixed in $2:1$ molar ratio.

The total pressure is $1.4\ atm$ and partial pressure of A_6 is $0.2\ atm$.

Question:

Calculate equilibrium partial pressures of A_2,

(A) 0.5

(B) 0.2

(C) 0.3

(D) 0.1

Q.16

Paragraph:

Following equilibria are established on mixing two gases A_2 and C.

(i) $3\ A_2(g) \rightleftharpoons A_6(g); K_p = 1.6\ atm^{-2}$

(ii) $A_2(g) + C(g) \rightleftharpoons A_2C(g)$

A_2 and C are mixed in $2:1$ molar ratio.

The total pressure is $1.4\ atm$ and partial pressure of A_6 is $0.2\ atm$.

@IITJEE_Advanced

Question:

Calculate equilibrium partial pressures of A_2C

(A) 0.5

(B) 0.2

(C) 0.3

(D) 0.1

Q.17

Paragraph:

Following equilibria are established on mixing two gases A_2 and C.

(i) $3\ A_2(g) \rightleftharpoons A_6(g); K_p = 1.6\ atm^{-2}$

(ii) $A_2(g) + C(g) \rightleftharpoons A_2C(g)$

A_2 and C are mixed in $2:1$ molar ratio.

The total pressure is $1.4\ atm$ and partial pressure of A_6 is $0.2\ atm$.

Question:

K_p for equation (ii) is

(A) $2.5\ atm^{-1}$

(B) $0.5\ atm^{-1}$

(C) $3.5\ atm^{-1}$

(D) $1.5\ atm^{-1}$

Q.18

Match the column -

Column I	**Column II**
(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(p) K_p is independent of P
(B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$	(q) $K_p = \frac{27\alpha^4P^2}{16(1-\alpha^2)^2}$
(C) $2ClF_3(g) \rightleftharpoons Cl_2(g) + 3F_2(g)$	(r) $K_p = \frac{\alpha^2P}{1-\alpha^2}$
(D) $NO(g) \rightleftharpoons N_2(g) + O_2(g)$	(s) $K_p = \frac{4\alpha^2P}{1-\alpha^2}$
	(t) $K_C < K_P$ (Here P is total pressure at equilibrium)

(A) A-r,t; B-s,t ; C- q,t ; D-p;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,r,s;

Q.19

@IITJEE_Advanced

At some temperature and under a pressure of 4 atm, PCl_5 is 10% dissociated. Calculate the pressure at which PCl_5 will be 20% dissociated, temperature remaining same.

Q.20

The degree of dissociation of I_2 molecule at $1000^\circ C$ and under 1.0 atmospheric pressure is 40% by volume. If the dissociation is reduced to 20% at the same temperature, then if the total equilibrium pressure on the gas is 4.57×10^x. Find x ?

Answers & Solutions

Q.1 Answer:

$4, 0.43, 0.93$

Solution:

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Let 1 mole each of ethyl alcohol, (C_2H_5OH) and acetic acid, (CH_3COOH) are mixed together as both are present in equimolar proportions. The reaction will be -

$C_2H_5OH + CH_3COOH \rightarrow CH_3COOC_2H_5 + H_2O$

The fraction of both consumed $= \frac{2}{3}$

$\therefore$ **No. of moles of** C_2H_5OH **and** CH_3COOH **left** $= 1 - \frac{2}{3} = \frac{1}{3}$

No. of moles of $CH_3COOC_2H_5$ **and** H_2O **at equillibrium** $= \frac{2}{3}$

$\therefore$ **The equilibrium constant for the reaction,** $(K_c) = \frac{[H_2O][CH_3COOC_2H_5]}{[C_2H_5OH][CH_3COOH]}$

$\Rightarrow K_c = \frac{\frac{2}{3} \times \frac{2}{3}}{\frac{1}{3} \times \frac{1}{3}} = \frac{\left(\frac{4}{9}\right)}{\left(\frac{1}{9}\right)} = 4$

(a) Let x moles of ester is formed from 1 mole of acid and 0.5 mole of alcohol, then

$K_C = \frac{x^2}{(1-x)(0.5-x)} \frac{x^2}{(1-x)(0.5-x)} = 4; x = 0.43$

(b) $K_C = \frac{x^2}{(1-x)(4-x)}$ **or** $4 = \frac{x^2}{(1-x)(4-x)}; x = 0.93$

Q.2 Answer:

$27/16 = 1.7 \text{ mol}^2\ell^{-2}$

Solution:

Let α be the degree of dissociation:

Hence we have :

$$\begin{array}{ccccc} 2NH_3(g) & \rightleftharpoons & N_2(g) & + & 3H_2(g) \\ 2 & & 0 & & 0 \\ 2-2\alpha & & \alpha & & 3\alpha \end{array}$$

$\text{At equilibrium : } [NH_3] = 2 - 2\alpha = 1 \text{ or } \alpha = \frac{1}{2}$

$[N_2] = \frac{1}{2}, [H_2] = \frac{3}{2}, [NH_3] = 1$

$\because \quad K = \frac{[N_2][H_2]^3}{[NH_3]^2} = \frac{\frac{1}{2} \times \frac{3}{2} \times \frac{3}{2} \times \frac{3}{2}}{1 \times 1} = \frac{27}{16} = 1.7 \text{ mol}^2\ell^{-2}$

Q.3 Answer:

@IITJEE_Advanced

$0.5 \times (1.5)^3 \text{ atm}^2$

Solution:

$3 \text{ mole } N_2, 1 \text{ mole } N_2$

$\text{total pressure } = 4 \text{ atm}$

$\Rightarrow \quad P_{H_2} = 3 \text{ atm}, P_{N_2} = \text{ atm}$

$$\begin{array}{cccccc} & N_2(g) & + & 3H_2(g) & \rightleftharpoons & 2NH_3(g) \\ \text{at eq.} & 1-x & & 2-3x & & 2x \end{array}$$

$P_{\text{total}} = 3 = 1 - x + 3 - 3x + 2x$

$\Rightarrow \quad 3 = 4 - 2x \Rightarrow x = \dfrac{1}{2}$

$P_{N_2} = 1/2; P_{H_2} = 3/2; P_{NH_3} \text{ (eq.)} = 1$

$$K_p = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^2} = \frac{(1)^2}{\frac{1}{2} \times \frac{27}{8}} = \frac{16}{27} = \frac{1}{0.5 \times (1.5)^3}$$

K_p **for dissociation of** $NH_3 = \frac{27}{16} = 0.5 \times (1.5)^3$

Q.4 Answer:

600 mm

Solution:

$\text{Moles of } H_2 = \frac{4 g H H_2}{2 \text{ g/mol}} = 2 \text{ mol} H_2$

$1 \text{ mole } = 6.023 \times 10^{23} \text{ molecules of } Cl_2$

let x **mole** $= 9 \times 10^{23}$ **molecules**

$x = \frac{9}{6.023} \times \frac{10^{23}}{10^{23}} = 1.494 \text{ mol} Cl_2$

$$\begin{array}{ccccc} H_2 & + & Cl_2 & \rightarrow & 2HCl \\ 2 & & 1.494 & & 0 \end{array}$$

As, Cl_2 **is limiting**

Moles HCl **formed** $= 2 \times 1.494 = 2.9885$ **moles**

Moles of H_2 **left=2** $-1.494 \text{ mol} = 0.506$ **moles**

$X_{HCl} = \frac{n_{HCl}}{nHCl^{nCl} + nH_H} = \frac{2.9885}{2.9885 + 0.506} = \frac{2.9885}{3.4945} = 0.855$

$P_{HCl} = X_{HCl} \times P_T$

$P_{HCl} = 0.855 \times 700 = 598.64 = 600 \text{ mm}$

Q.5 Answer:

$SrCl_2 \cdot 2H_2O$

Solution:

$(P_{H_2O})_1 = \left(5 \times 10^{12}\right)^{\frac{1}{4}} = (5)^{1/4} \times 10^3$

$(P_{H_2O})_2 = \left(2.43 \times 10^{13}\right)^{\frac{1}{5}} = 3 \times 10^3$

$(P_{H_2O})_3 = \left(1.204 \times 10^{27}\right)^{\frac{1}{10}} = 2 \times 10^3$

@IITJEE_Advanced

as $(P_{H_2O})_1$ is least hence $SrCl_2.2H_2O$ will act as good dehydrating agent

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g) K_p = 27 \times 10^8 P_0^2$

Q.6 Answer:

7.0×10^{-16}

Solution:

$2NO \rightleftharpoons N_2 + O_2; K_1 = 4 \times 10^{30}$

$$N_2 + O_2 \rightleftharpoons 2NO; \quad K' = \frac{1}{K_1} = \frac{1}{4 \times 10^{30}}$$

Divide by 2 ,

(i) $\frac{1}{2}\ N_2 + \frac{1}{2} O_2 \rightleftharpoons NO; \quad K'' = \sqrt{K'} = \frac{1}{2 \times 10^5}$

(ii) $NO + \frac{1}{2} Br_2$ **(g)** $\rightleftharpoons NOBr; K_2 = 1.4\ mol^{-1/2}\ L^{1/2}$

On adding (i) \& (ii)

$$\frac{1}{2}\ N_2(\ g) + \frac{1}{2} O_2(\ g) + \frac{1}{2} Br_2(\ g) \rightleftharpoons NOBr(g)$$
$$K = K'' \times K_2 = \frac{1.4}{2 \times 10^5} = 7.0 \times 10^{-16}$$

Q.7 Answer:

25%

Solution:

Let no. of moles of $NH_3 = n_1$ **and** $N_2H_4 = n_2$ **in the initial mixture**

Total no. of moles initially $= n_1 + n_2$

Total no. of moles after decomposition $= 2n_1 + 3n_2$

$\therefore 0.5 \times v = (n_1 + n_2) R \times 300$

$4.5 \times v = (2n_1 + 3n_2) R \times 1200$

$\frac{2n_1 + 3n_2}{n_1 + n_2} = \frac{9}{4}, n_1 = 3n_2$

$\frac{9n_2}{n_1 + n_2} = \frac{9}{4}$ $\quad \begin{cases} \frac{n_2}{n_1 + n_2} = \frac{1}{4} \\ \therefore \%\ \text{mole of } N_2H_4 = 25\% \end{cases}$

Q.8 Answer:

Forward if K_C for the reaction is 14

Solution:

$A + B \rightleftharpoons 3C$

Given, $[A] = 1/3, [\ B] = 2/3, [C] = 4/3$

@IITJEE_Advanced

$$Q = \frac{[C]^3}{[A][B]} = \frac{64}{6} = 10.66$$

If $K_C < Q$, equilibrium shifts into backward direction.

$K_C > Q$, equilibrium shifts into forward direction.

Q.9 Answer:

increasing temperature favours forward reaction

Solution:

$NaNO_3(s) \rightleftharpoons NaNO_2(s) + O_2(g)$

$\Delta H > 0$ **endothermic, High temperature, forward, low pressure favours it High pressure $\rightarrow$ Reverse reaction**

Q.10 Answer:

$[AB_3]_{eq} > [B_3C_2]_{eq}$

Solution:

Let reactions is started with a mole of AB_2 and b mole of B_2C.

$$\begin{array}{cccc} AB_2(g) + B_2C(g) \rightarrow & AB_3(g) + & BC(g) & \\ a \quad\quad b & 0 \quad\quad & 0 & \\ a-x \quad b-x-y & x \quad\quad & x-y & \\ BC(g) \quad + & B_2C(g) \quad \rightarrow & B_3C_2(g) & \\ x-y \quad\quad & b-x-y \quad\quad & y & \end{array}$$

As $y > x$

Clearly $[AB_3]_{eq} > [B_3C_2]_{eq}$ and $[AB_3]_{eq} > [BC]_{eq}$

Q.11 Answer:

Total pressure is 1.5 times that of partial pressure of NH_3.

Solution:

$$NH_2COONH_4 \underset{(Solid)}{\rightleftharpoons} 2NH_3 \underset{(g)}{+} CO_2 \underset{(g)}{}$$

@IITJEE_Advanced

What ever may be extent of dissociation NH_3 and CO_2 are formed in $2:1$ mole ratio.

$\therefore \quad P_{NH_3} = 2x, P_{CO_2} = x.$

Total pressure $= 3x$.

$K_p = P_{NH_3}^2 \cdot P_{CO_2}$

$= (2x)^2 \cdot x = 4x^3$

$\therefore K_p = 4P_{CO_2}^3$

$\frac{P_{potal}}{P_{NHL}} = \frac{3x}{2x} = 1.5$

Q.12 Answer:

$3/4$

Solution:

$2\,A_2(g) \quad \rightleftharpoons \quad A_4(g)$

$3p - x - y \qquad\quad x/4$

$A_2 \quad + \quad 2C \quad \rightleftharpoons \quad A_2C_2$

$3p - x - y \qquad p - 2y \qquad\quad y - z$

$A_2C_2 \quad \rightleftharpoons \quad 2AC$

$y - z \qquad\quad 2z$

$\frac{P_{A_4}}{P_{A_2}^2} = K_{P_1} \Rightarrow (P_{A_2})^2 = \frac{P_{A_4}}{K_{P_1}} = \frac{1/2}{2/81} = \frac{81}{4}$

$\Rightarrow \quad P_{A_2} = \frac{9}{2}$ atm

$\Rightarrow 3p - x - y = 9/2$

$\frac{x}{2} = \frac{1}{2}$

$\Rightarrow \quad x = 1$ atm

Also given $2z = 1/2$

$z = 1/4$ atm

$P_{total} = 3p - x - y + x/2 + p - 2y + y - z + 2z = 4p - x/2 - 2y + z$

$= 4 \times p - \frac{1}{2} - 2 \times \frac{1}{2} + \frac{1}{4} = \frac{27}{4} atm \Rightarrow P = 2atm$

$3p - x - y = \frac{9}{2}; \quad 6 - \frac{1}{2} - y = \frac{9}{2}$

$y = 1$ atm

$P_{A_2C_2} = y - z = 1 - \frac{1}{3} = \frac{3}{4}$ atm.

$\frac{n_{A_2}}{n_{AC}} = \frac{P_{A_2}}{P_{AC}} = \frac{3P - x - y}{2z} = \frac{9/2}{1/2} = 9$

$K_p = \frac{P_{A_2C_2}}{P_{AC}^2} = \frac{3/4}{(1/2)^2} = 3$

Q.13 Answer:

9

Solution:

$2\,A_2(g) \quad \rightleftharpoons \quad A_4(g)$

$3p - x - y \qquad\quad x/4$

$A_2 \quad + \quad 2C \quad \rightleftharpoons \quad A_2C_2$

$3p - x - y \qquad p - 2y \qquad\quad y - z$

@IITJEE_Advanced

$A_2C_2 \rightleftharpoons 2AC$

$y-z \quad 2z$

$\frac{P_{A_4}}{P_{A_2}^2} = K_{P_1} \Rightarrow (P_{A_2})^2 = \frac{P_{A_4}}{K_{P_1}} = \frac{1/2}{2/81} = \frac{81}{4}$

$\Rightarrow \quad P_{A_2} = \frac{9}{2}$ atm

$\Rightarrow 3p - x - y = 9/2$

$\frac{x}{2} = \frac{1}{2}$

$\Rightarrow \quad x = 1$ atm

Also given $2z = 1/2$

$z = 1/4$ atm

$P_{\text{total}} = 3p - x - y + x/2 + p - 2y + y - z + 2z = 4p - x/2 - 2y + z$

$= 4 \times p - \frac{1}{2} - 2 \times \frac{1}{2} + \frac{1}{4} = \frac{27}{4} atm \Rightarrow P = 2atm$

$3p - x - y = \frac{9}{2}; \quad 6 - \frac{1}{2} - y = \frac{9}{2}$

$y = 1$ atm

$P_{A_2C_2} = y - z = 1 - \frac{1}{3} = \frac{3}{4}$ atm.

$\frac{n_{A_2}}{n_{AC}} = \frac{P_{A_2}}{P_{AC}} = \frac{3P-x-y}{2z} = \frac{9/2}{1/2} = 9$

$K_p = \frac{P_{A_2C_2}}{P_{AC}^2} = \frac{3/4}{(1/2)^2} = 3$

Q.14 Answer:

3

Solution:

$2\,A_2(g) \rightleftharpoons A_4(g)$

$3p - x - y \quad x/4$

$A_2 + 2C \rightleftharpoons A_2C_2$

$3p - x - y \quad p - 2y \quad y - z$

$A_2C_2 \rightleftharpoons 2AC$

$y - z \quad 2z$

$\frac{P_{A_4}}{P_{A_2}^2} = K_{P_1} \Rightarrow (P_{A_2})^2 = \frac{P_{A_4}}{K_{P_1}} = \frac{1/2}{2/81} = \frac{81}{4}$

$\Rightarrow \quad P_{A_2} = \frac{9}{2}$ atm

$\Rightarrow 3p - x - y = 9/2$

$\frac{x}{2} = \frac{1}{2}$

$\Rightarrow \quad x = 1$ atm

Also given $2z = 1/2$

$z = 1/4$ atm

$P_{\text{total}} = 3p - x - y + x/2 + p - 2y + y - z + 2z = 4p - x/2 - 2y + z$

$= 4 \times p - \frac{1}{2} - 2 \times \frac{1}{2} + \frac{1}{4} = \frac{27}{4} atm \Rightarrow P = 2atm$

$3p - x - y = \frac{9}{2}; \quad 6 - \frac{1}{2} - y = \frac{9}{2}$

$y = 1$ atm

$P_{A_2C_2} = y - z = 1 - \frac{1}{3} = \frac{3}{4}$ atm.

@IITJEE_Advanced

$\frac{n_{A_2}}{n_{AC}}=\frac{P_{A_2}}{P_{AC}}=\frac{3P-x-y}{2z}=\frac{9/2}{1/2}=9$

$K_p=\frac{P_{A_2C_2}}{P_{AC}^2}=\frac{3/4}{(1/2)^2}=3$

Q.15 Answer:

0.5

Solution:

$3\ A_2(\ g) \rightleftharpoons A_6(\ g) K_p=1.6\ atm^{-2}$

$K_p=1.6=\frac{P'_{A_s}}{\left(P_{A_2}\right)^3}$

$P_{A_2}=\sqrt[3]{\frac{0.2}{1.6}}=0.5\ atm$

Let the partial pressure of A_2 and C before the start of reaction be a and a $/2$ atm respectively

$A_2(\ g)+C(g) \rightleftharpoons A_2C(g)$

If x moles of A_2 reacts to form x moles of A_2C, then

$1.4=0.5+0.2+P_{A_2C}+P_C; \quad P_{A_2C}+P_C=0.7$

$a/2-x+x=0.7; \quad a=1.4\ atm.$

Now, partial pressure of A_2 consumed in Ist reaction $=3\times0.2=0.6\ atm$

Partial pressure of A_2C consumed in IInd reaction $=1.4-3\times0.2-0.5=0.3\ atm$

$P_{A_2C}=0.3\ atm$

$P_c=0.7-0.3=0.4\ atm.$

$K_p=\frac{0.3}{0.4\times0.5}=1.5\ atm^{-1}$

Q.16 Answer:

0.3

Solution:

$3A_2(g) \rightleftharpoons A_6(g)$

$K_p=1.6=P_{A_2}^3=0.5\ atm$

Let the partial pressure of A_2 and C before the start of reaction be a and a $/2$ atm respectively

$A_2(g)+C(g) \rightleftharpoons A_2C(g)$

If x moles of A_2 reacts to form x moles of A_2C, then

$1.4=0.5+0.2+P_{A_2C}+P_C; \quad P_{A_2C}+P_C=0.7$

$a/2-x+x=0.7; \quad a=1.4\ atm.$

Now, partial pressure of A_2 consumed in Ist reaction $=3\times0.2=0.6\ atm$

Partial pressure of A_2C consumed in IInd reaction $=1.4-3\times0.2-0.5=0.3\ atm$ **replace old solution by this**

$P_{A_2C}=0.3\ atm$

$P_c=0.7-0.3=0.4\ atm.$

$K_p=\frac{0.3}{0.4\times0.5}=1.5\ atm^{-1}$

Q.17 Answer:

$1.5\ atm^{-1}$

@IITJEE_Advanced

Solution:

$$3\ A_2(g) \rightleftharpoons A_6(g)$$
$$K_p = 1.6 = P_{A_2}^3 = 0.5\ atm$$

Let the partial pressure of A_2 and C before the start of reaction be a and a $/2$ atm respectively

$$A_2(g) + C(g) \rightleftharpoons A_2C(g)$$

If x moles of A_2 reacts to form x moles of A_2C, then

$$1.4 = 0.5 + 0.2 + P_{A_2C} + P_C; \quad P_{A_2C} + P_C = 0.7$$
$$a/2 - x + x = 0.7; \quad a = 1.4\ atm.$$

Now, partial pressure of A_2 consumed in Ist reaction $= 3 \times 0.2 = 0.6$ atm
Partial pressure of A_2C consumed in IInd reaction $= 1.4 - 3 \times 0.2 - 0.5 = 0.3$ atm

$$P_{A_2C} = 0.3\ atm$$
$$P_c = 0.7 - 0.3 = 0.4\ atm.$$
$$K_p = \frac{0.3}{0.4 \times 0.5} = 1.5\ atm^{-1}$$

Q.18 Answer:

A-r,t; B-s,t ; C- q,t ; D-p;

Solution:

(A)-rt, (B)-st, (C) - qt, (D) - p

(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$K_P = \frac{P_{PCl_3} \times P_{PCl_2}}{P_{PCl_5}} = \frac{\frac{\alpha P}{1+\alpha} \times \frac{\alpha P}{1+\alpha}}{\frac{(1-\alpha)P}{1+\alpha}} = \frac{\alpha^2}{1-\alpha^2}P$$

(B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$1-\alpha \qquad 2\alpha$

$$K_P = \frac{\frac{4\alpha^2}{(1+\alpha)^2} \times P^2}{\frac{1-\alpha}{1+\alpha} \times P} = \frac{4\alpha^2 P}{1-\alpha^2}$$

(C) $2ClF_3(g) \rightleftharpoons Cl_2(g) + 3F_2(g)$

$1-\alpha \qquad \frac{\alpha}{2} \qquad \frac{3\alpha}{2}$

$$\text{Total} = 1 - \alpha + 2\alpha = 1 + \alpha$$
$$= \frac{\left(\frac{3\alpha}{2} \times \frac{1}{1+\alpha}\right)^3 \left(\frac{\frac{\alpha}{2}}{1+\alpha}\right) \times P^2}{\left(\frac{1-\alpha}{1+\alpha}\right)^2 \times P^2} = \frac{27}{16} \times \frac{\alpha^4}{(1+\alpha)^2} \times \frac{1}{(1-\alpha)^2} \times P^2$$
$$K_P = \frac{27\alpha^4 P^2}{16(1-\alpha^2)^2}$$

(D) $\underset{1-\alpha}{NO(g)} \rightleftharpoons \underset{\frac{\alpha}{2}}{\frac{1}{2}N_2(g)} + \underset{\frac{\alpha}{2}}{\frac{1}{2}O_2(g)}$

@IITJEE_Advanced

$K_P = \left(\frac{\alpha}{2} \times P\right)^{1/2} \left(\frac{\alpha}{2} \times P\right)^{1/2}$

hence independent of pressure

Q.19 Answer:

—

Solution:

$PCl_5 \longrightarrow PCl + Cl_2$

$P - \frac{P}{10} \quad \frac{P}{10} \frac{P}{10}$

$P_{total} = P + \frac{P}{10} = \frac{11P}{90} = 4$

$\Rightarrow P = \frac{40}{11}$

$K_P = \frac{P^2/100}{9^P/10} = \frac{P}{90} = \frac{4}{99}$

If 20% dissociation

$PCl_5 \longrightarrow PCl + Cl_2$

$P - \frac{P}{5} \quad \frac{P}{5} \quad \frac{P}{5}$

$\frac{4}{99} = \frac{\left(P^1/5\right)^2}{(4P^1/5)} \Rightarrow P^1 = \frac{80}{99}$

$P_{total} = \frac{6P^1}{5} = \frac{6}{5} \times \frac{80}{99}$

$= 0.97$

Q.20 Answer:

—

Solution:

$K_p = \frac{4\alpha^2 P}{1-\alpha^2}$

$\alpha = 0.4, \quad p = 1 \text{ atm}$

$\frac{4\times0.16}{0.84} = \frac{4\times0.04}{0.96} P_{new}$

$P_{new} = 4.57 \text{ atm}$

@IITJEE_Advanced